MIL-C-5824A
11 MARCH 1952

Superseding
MIL-C-5824(USAF)
17 August 1950

MILITARY SPECIFICATION

COURSE INDICATOR ID-250A/ARN

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers one type of course indicator which indicates magnetic bearing of two radio stations and the magnetic heading of the aircraft, designated ID-250A/ARN.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications.-

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-C-5015	Connectors; Electrical
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-D-5028	Drawings and Data Lists; Preparation of (For Engines, Accessories and Other Auxiliary Equipment)
MIL-E-5272(USAF)	Environmental Testing, Aeronautical and Associated Equipment, (General Specification for)
MIL-I-6181	Interference Limits and Tests; Aircraft Electrical and Electronic Equipment
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-S-7742	Screw Threads, Standard, Aeronautical

Air Force-Navy Aeronautical

AN-QQ-A-696	Anodic-Films; Corrosion-Protective (For) Aluminum Alloys
AN-L-1	Luminescent Material; Fluorescent
AN-M-13	Marking, Labeling, and Tagging, Aircraft Aeronautical Equipment, Accessories, Parts, Materials and Supplies

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2.1.2 Other Publications.-

Air Force-Navy Aeronautical Bulletin

No. 143 Specifications and Standards; Use of

2.1.3 Drawings.-

Military Standards (Sheet)

MS28027 Course Indicator ID-250A/ARN

Air Force-Navy Aeronautical Standard Drawing

AND10400 Numerals and Letters - Aircraft Instrument Dial, Standard Form of

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.-

3.1.1 Metals.- Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life. Materials contained within hermetically sealed enclosures are considered to be suitably protected from corrosion. Requirements specified for fungicidal and corrosion-protective treatment, anodizing of aluminum-alloy parts, and cadmium plating of steel parts are not applicable for parts within hermetically sealed instruments.

3.1.2 Nonmagnetic Materials.- Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.1.3 Fungus-Proof Materials.- Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent, as approved by the procuring agency.

3.1.4 Protective Treatment.- When materials are used in the construction of the indicator that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.1.5 Selection of Materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.1.5.1 AN or JAN Standard Parts.- AN or JAN Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the AN or JAN Standard parts without alteration, and provided the corresponding AN or JAN part

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numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding AN or JAN Standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.2 Design.- The indicator shall be designed to receive synchronous electrical signals from a magnetic compass system and two radio compass systems, combine the magnetic compass data with the radio compass data and indicate magnetic heading of the aircraft on a rotating dial and magnetic bearing of two radio stations under two pointers.

3.3 Construction.- The indicator shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.3.1 Indicator Case.- The case shall conform to Standard MS28027. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The design of the indicator shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures, unless they are approved by the procuring agency. The hermetic seal shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be submitted.

3.3.1.1 The case shall be constructed of a suitable material applicable to hermetic sealing, and the material shall be uniform in texture with a smooth external surface.

3.3.1.2 The case shall be filled with a mixture of 10 percent helium and 90 percent nitrogen and should contain not more than 0.006 milligram of water vapor per liter (dew point -65°C) at approximately one atmosphere filling pressure.

3.3.2 Cover Glass.- The cover glass shall be suitable for hermetic sealing and shall be clear, flat, and free from flaws which interfere with normal reading of the indicator. The distance between the cover glass and the dial shall be the minimum practicable and shall not exceed 0.1875 inch.

3.3.3 Dial.- The dial shall conform to figure 1. The style and proportions of numerals and letters placed on the dial shall conform to Drawing AND10400. Numerals shall distinctly indicate the graduation to which each applies.

3.3.4 Pointers.- The pointers shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. They shall be in accordance with figure 2. The pointers shall be firmly attached to the mechanism, and yet be readily adjustable.

3.3.4.1 The front pointer shall be the radio compass pointer and shall be labeled with the numeral "1."

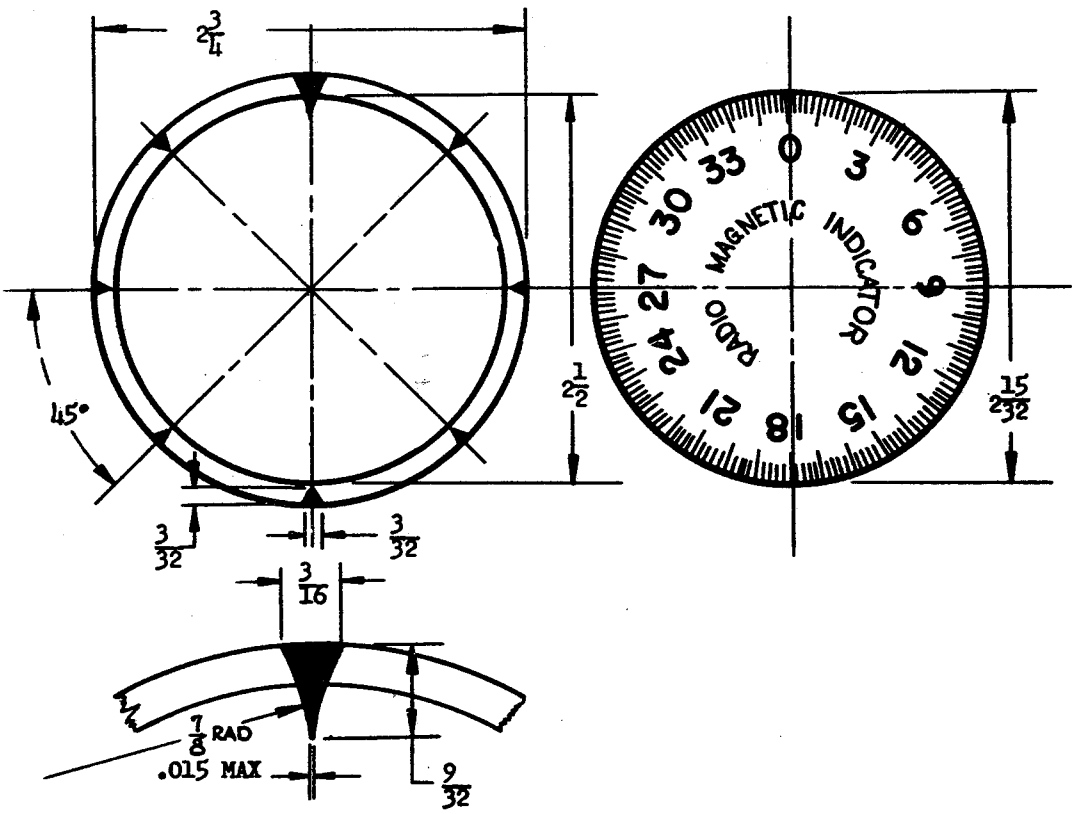
3.3.4.2 The rear pointer shall be the omni-directional pointer and shall be labeled with the numeral "2."

3.4 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.5 Screw Threads.- Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to Specification MIL-S-7742.

3.6 Electrical Connectors.- The electrical connector used in the construction of the indicator shall be in accordance with Specification MIL-C-5015, except in hermetically sealed instruments where a receptable designed for hermetic sealing may be used.

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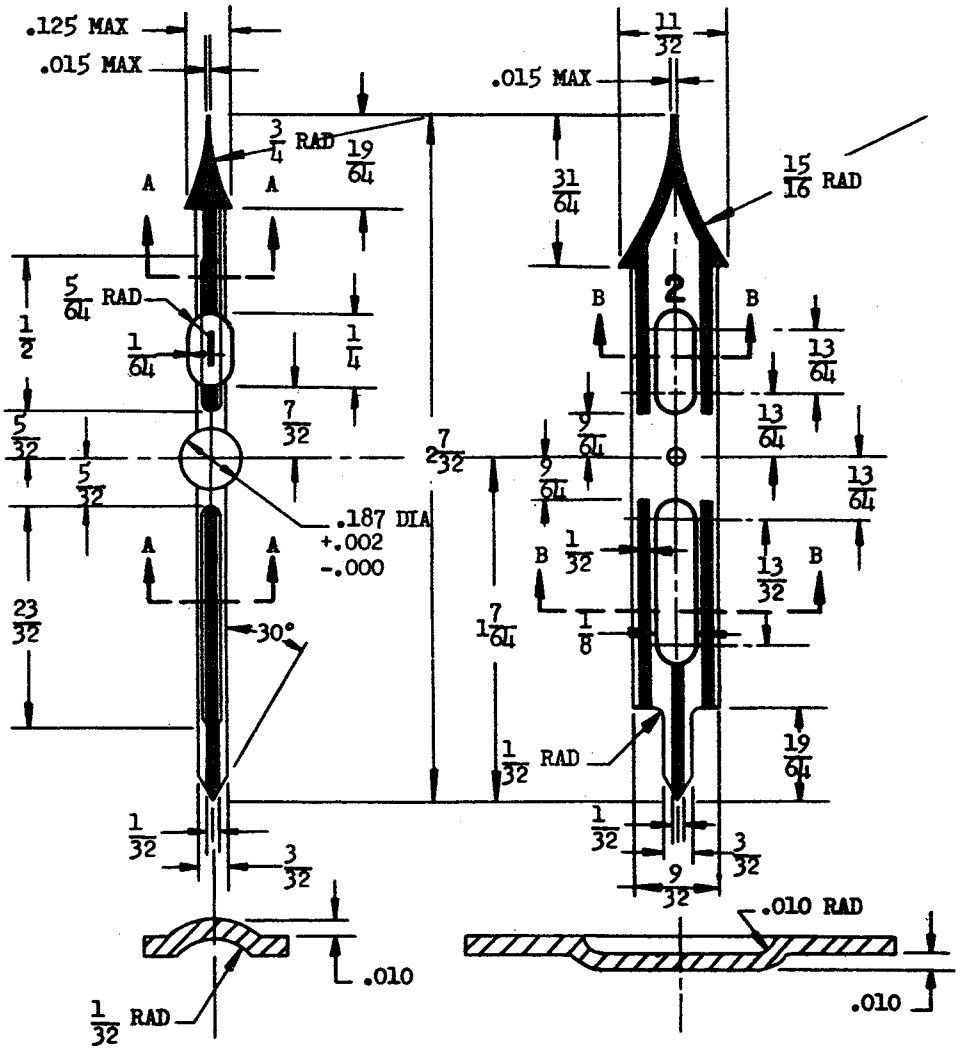


POINTER ON RING

MARKINGS	HEIGHT OR LENGTH ±0.010	WIDTH OF LINE OR GRADUATIONS ±0.005	MATERIAL OR FINISH
NUMERALS	0.187	0.015	FLUORESCENT LUMINESCENT SPEC AN-L-1
2-DEGREE GRADUATIONS	0.125	0.010	
10-DEGREE GRADUATIONS	0.187	0.020	
FUDICIAL MARKER	- - -	- - -	
45-DEGREE TRIANGLE INDEX MARKERS	- - -	- - -	
LETTERING "RADIO MAGNETIC INDICATOR"	0.125	0.015	DURABLE DULL BLACK
BACKGROUND OF SCALE	- - -	- - -	
UNSHADED PORTION OF POINTER	- - -	- - -	

FIGURE 1. Dial

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MARKINGS	HEIGHT OR LENGTH ±0.010	WIDTH OF LINE OR GRADUATIONS ±0.005	MATERIAL OR FINISH
POINTERS - SHADED PORTION NUMERALS "1" AND "2"	- - - 0.156	- - - 0.015	FLUORESCENT-LUMINESCENT
POINTERS UNSHADED PORTION	- - -	- - -	DURABLE DULL BLACK

FIGURE 2. Pointer

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3.7 Wiring Diagram.- The wiring diagram for the indicator shall be as shown on figure 3.

3.8 Weight.- The weight of the indicator completely assembled, less mounting nuts and mounting screws, shall not exceed 3 pounds.

3.9 Finishes and Protective Coatings.-

3.9.1 Aluminum-Alloy Parts.- Aluminum-alloy parts shall be covered with an anodic film conforming to Specification AN-QQ-A-696, except as follows.

3.9.1.1 Dials, small holes, and case inserts need not be anodized.

3.9.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.9.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with Specification MIL-C-5541 may be used in lieu of anodizing.

3.9.1.4 Castings containing non-aluminum-alloy integral inserts may be treated with a chemical film in accordance with Specification MIL-C-5541 in lieu of anodizing.

3.9.1.5 When abrasion resistance is a factor, chemical films in accordance with Specification MIL-C-5541 shall not be used in lieu of anodizing.

3.9.2 Steel Parts.- Steel parts shall be cadmium-plated, where practicable, in accordance with Specification QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.10 Performance.- The indicator shall meet the performance requirements specified in Section 4, when subjected to the applicable tests.

3.11 Identification of Product.-

3.11.1 Nameplate.- A nameplate, permanently and legibly filled in with the information listed below shall be securely attached to the indicator. The information marked in the spaces provided on the nameplate shall be in accordance with Specification AN-M-13.

COURSE INDICATOR ID-250A/ARN
Specification MIL-C-5824A
MS Part No.
Stock No. (USAF or Navy, as applicable)
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or Trade-Mark
U S Property

3.12 Workmanship.-

3.12.1 General.- The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

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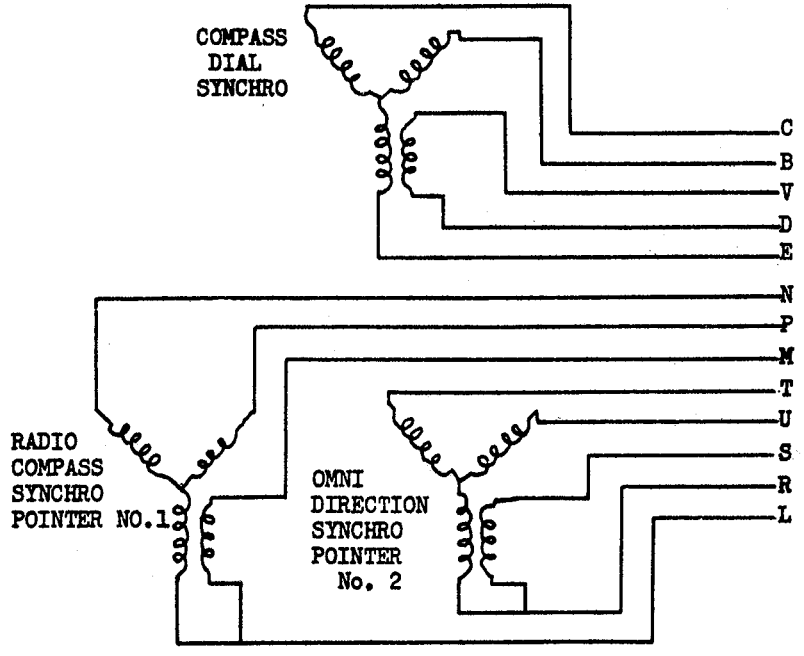


FIGURE 3.

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3.12.2 Dimensions.- Dimensions and tolerances not specified shall be as close as is consistent with best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.12.3 Screw Assemblies.- Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.12.4 Riveting.- Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.12.5 Gears.- Gear assemblies shall be properly alined and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.12.6 Cleaning.- The indicator shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of Tests.- The inspection and testing of indicators shall be classified as follows:

(a) Qualification tests: Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

(b) Inspection tests: Inspection tests are those tests accomplished on indicators manufactured and submitted for acceptance under contract.

4.2 Test Conditions.-

4.2.1 Standard Atmospheric Conditions.- Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.2.2 Attitude.- Unless otherwise specified, the indicator shall be tested in a normal operating position.

4.2.3 Tapping.- Unless otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken.

4.2.4 Vibration.- The vibration equipment shall be in accordance with Specification MIL-E-5272(USAF), procedure III.

4.2.5 Standard Test Transmitter.- The standard test transmitter shall be an Eclipse-Pioneer type AY 201-1, or equal, high precision transmitting synchro. The rotor of the synchro shall be settable, rotatable, and readable through 360 degrees of rotation to 1 ±0.1 degree. The following procedure shall be used to calibrate the test transmitter.

4.2.5.1 The leads of rotor "Hot" and "Cold" shall be designated at will.

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4.2.5.2 One of the stator leads shall be selected, designated, and labeled "Z," and tied to the "Cold" rotor lead.

4.2.5.3 A voltage of 26V, 400 cycles shall be applied to the "Hot" lead, and the "Cold" and "Z" leads shall be grounded. The rotor or stator shall be rotated until the output of the remaining leads is at a "null" or minimum voltage.

4.2.5.4 The voltage between the "Hot" lead and the remaining stator leads strapped shall be measured. If this voltage is less than the excitation voltage of the rotor, the synchro rotor is at "low null" or the 180-degree position. If the voltage is greater than the rotor excitation voltage, the synchro rotor is at "high null" or the zero-degree position.

4.2.5.5 The rotor shall be placed at the "high null" or zero-degree position. A voltmeter shall then be connected across one of the remaining stator leads and the ground, with the second lead open. When the synchro rotor is turned clockwise, the voltage shall increase before decreasing if the "X" stator lead is connected. The voltage shall decrease before increasing if the "Y" stator lead is connected. All the leads of the synchro shall be labeled in accordance with the results of the above tests.

4.3 Qualification Tests.-

4.3.1 Sampling Instructions.- Qualification test samples shall consist of five indicators. Samples shall be identified as required and forwarded to the agency designated in the Letter of Authorization from the qualifying agency. (See paragraph 6.2.)

4.3.2 Tests.- The Qualification tests shall consist of all the tests described under Inspection tests, (subject to the test conditions specified in paragraph 4.2) and, in addition, the ones described herein.

4.3.2.1 Radio Noise Interference.- The radio noise interference test shall be in accordance with Specification MIL-I-6181.

4.3.2.2 Environmental Tests.- The following tests shall be in accordance with Specification MIL-E-5272(USAF) and as specified herein. Upon completion of each test, the compass shall be subjected to the Individual tests specified herein.

4.3.2.2.1 High Temperature: Procedure I.- The indicator shall be operated during the test.

4.3.2.2.2 Low Temperature: Procedure III.- The indicator shall be operated at the end of the tests. Allowable friction error of the pointer shall not exceed 2 degrees and the dial 3 degrees.

4.3.2.2.3 Humidity: Procedure I.- Upon removal from the humidity chamber, the indicator shall be operated at room temperature for 36 hours, at which time the indicator shall operate without any sticking of pointers or dial.

4.3.2.2.4 High Altitude: Procedure III.- The indicator shall be operated during this test. Upon removal from the altitude chamber, the indicator shall be tested for friction error.

4.3.2.2.5 Mildew Resistance: Procedure I.- Upon completion of this test, the indicator shall be examined and operated to determine if any permanent damage was incurred. Permanent damage shall be cause for rejection.

4.3.2.2.6 Salt Spray: Procedure I.- Upon removal from the salt spray chamber, the indicator shall be operated at room temperature and tested for friction error.

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4.4 Inspection Tests.- The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.4.1 Individual Tests.- Each indicator shall be subjected to the following tests.

4.4.1.1 Examination of Product.- Each indicator shall be inspected to determine compliance with this specification with respect to material and workmanship.

4.4.1.2 Electrical Zero and Rotation Test.- The test transmitter shall be connected to the indicator, in turn, as follows:

Test Transmitter	Indicator Dial Test	No. 1 Pointer Test	No. 2 Pointer Test
"Hot"	V	M	S
"Cold"	D	L	R
"X"	B	N	T
"Y"	C	P	U
"Z"	E	L	R

4.4.1.2.1 When the test transmitter needle and rotor are at zero-degree position, the dial of the indicator shall indicate zero degrees, the numbers 1 and 2 pointers shall indicate 180 degrees. When the test transmitter rotor is rotated clockwise with increasing positive indication, the dial and pointers of the indicator shall rotate with increasing positive indication. When the test transmitter rotor is rotated counterclockwise with decreasing positive indication, the dial and pointers of the indicator shall rotate with decreasing positive indication.

4.4.1.3 Scale Error.- The standard test transmitter shall be again connected in succession as indicated in the Electrical Zero and Rotation test paragraph. The standard test transmitter shall be turned successively to headings of 0, 30, 60, 90, 120, 150, 180, 210, 240, 270, 300, 330, and 360 degrees and the readings recorded. The difference between the indicator heading and the standard transmitter heading for each synchro shall be added algebraically and the algebraic sum divided by the number of headings. The result, allowing for the electrical zero, shall be the index error, and shall not exceed ±1/4 degree. Any one reading shall not differ from the correct heading by more than ±3/4 degree. The indicator shall be tapped before the reading is taken.

4.4.1.4 Friction.- The Scale Error test shall be repeated, except that the readings of the indicator shall be taken before and after tapping of each heading. The change in reading, as a result of tapping, is friction error. The maximum allowed friction error of the dial shall not exceed 2 degrees, the maximum allowed friction error of the pointers shall not exceed 1 degree.

4.4.1.5 Sealing.- The indicator shall be immersed in a suitable liquid, such as water. The absolute pressure of the air above the liquid shall then be reduced to approximately 1 inch Hg and maintained for 1 minute, or until air bubbles substantially cease to be given off by the liquid, whichever is longer. The absolute pressure shall then be increased to 2-1/2 inches Hg. Any bubbles coming from within the indicator case shall be considered as leakage and shall be cause for rejection. Bubbles which are the result of entrapped air on the various exterior parts of the case shall not be considered as a leak. A helium leak detector or other means of test, equal or superior in sensitivity to the immersion method specified above, may be used upon approval by the procuring agency.

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4.4.2 Sampling Tests.- Five indicators shall be selected at random from each lot of 500 or fraction thereof on the order and subjected to the following tests. These tests shall be in addition to the Individual tests specified herein. A lot shall consist of indicators manufactured under essentially the same conditions and from the same batch of materials.

4.4.2.1 Low Temperature Operation.- The indicator shall be subjected to a temperature of -54° ±2°C for a period of 4 hours. At the end of this period and while still at the low temperature, the indicator shall meet the requirements specified for the Individual tests, except that the tolerances shall be as follows.

4.4.2.1.1 Friction Error.- The friction error of the pointers shall not exceed 2 degrees and the dial 3 degrees.

4.4.2.1.2 After the indicator has returned to room temperature, it shall meet the tolerances specified for the Individual tests.

4.4.2.2 High Temperature Operation.- The indicator shall be placed in a chamber and maintained at a temperature of 71° ±2°C for a period of 4 hours. While the indicator is still at this temperature, it shall meet the requirements specified for the Individual tests. After the indicator has returned to room temperature, it shall meet the requirements specified for the Individual tests.

4.4.2.3 Vibration Error.- The indicator shall be mounted on the vibration stand so that a line joining the centers of the two lower mounting holes is in a horizontal plane and the plane of the mounting lugs in a vertical position. The indicator shall be operated by the standard test transmitter. The indicator shall be subject to vibration with an amplitude between 0.003 and 0.005 inch at frequencies of from 500 to 3,000 cycles per minute.

4.4.2.3.1 Pointer Vibration.- The maximum amplitude of the pointer's vibration and the pointer's variation shall not exceed 1/64 inch.

4.4.2.3.2 Accuracy.- The reading shall not differ at any time by more than 1/2 degree from the reading obtained when the indicator is at rest.

4.4.2.4 Magnetic Effect.- The indicator, first not operating, and second operating on rated voltage, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0, 45, 90, 135, 180, 225, and 315 degrees from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 gauss. The deflection of the compass at any of the specified positions shall not exceed 2 degrees.

4.4.2.5 Damping.- Two indicators shall be connected to the standard test transmitter with their pointers (or dials) displaced (within the limits of 45 to 90 degrees) in opposite directions from the locked standard data transmitter setting, and instantly energized from the alternating current bus at 35V, 400 cycles. The time limit for the pointer (or dial) oscillation, if present, to damp out shall not exceed 4 seconds. The dials of the two indicators, the No. 1 pointers of both indicators, and the No. 2 pointer of both indicators shall be successively subjected to this test.

4.4.3 Rejection and Retest.-

4.4.3.1 When tests are specified on a quantity of indicators that are selected as representative of a certain lot, and one or more of this number fails to meet the requirements, additional indicators of the lot represented shall be tested immediately to determine the cause of failure. Individual performance tests shall not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the indicator.

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4.4.3.2 Rejected indicators shall not be resubmitted for inspection without furnishing full particulars concerning previous rejection and measures taken to overcome the defects.

4.5 All parts, specimens, or assemblies destroyed in making tests required by this specification and/or drawings, to determine compliance with the specification and/or drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging and Packing.- Each indicator shall be packaged and packed for shipment in accordance with group X of Specification MIL-P-6064, in a type II, class I container.

5.3 Marking and Labeling.- Marking shall be as specified in Specification MIL-P-6064. In addition, each container shall be marked with the following information:

COURSE INDICATOR ID-250A/ARN
Specification MIL-C-5824A
MS Part No.
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or Trade-Mark
Name of Contractor (if different from manufacturer)
Date of Manufacture
Quantity

IF IN STORAGE AFTER (DATE)*, THE INDICATOR SHALL BE TESTED AND INSPECTED BEFORE ISSUE.

* NOTE: This date shall be 36 months from the date of inspection and shall be inserted by the Inspector at the time of acceptance.

5.3.1 Precautionary Marking.- The following precautionary marking shall be included on the shipping container:

"FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE"

6. NOTES

6.1 Ordering Data.- Procurement documents should specify the title, number, and date of this specification.

6.2 Provisions for Qualification Tests.- The right is reserved to reject any bids on items which have not been subjected to the required tests and found satisfactory. The attention of the manufacturers is called to this provision and they are urged to request authorization for tests of the items which they propose to offer to the Air Force or Navy under this specification. Requests for authorization of tests, together with certified

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test reports showing conformance with all the requirements of this specification and the manufacturer's assembly and detail drawings, and for information as to the marking and forwarding of samples should be addressed to the Commanding General, Wright Air Development Center, Wright-Patterson Air Force Base, Dayton, Ohio, the qualifying agency, with a copy to the Bureau of Aeronautics, Navy Department, Washington 25, D. C.

6.2.1 It is to be understood that upon receipt of the Letter of Authorization, samples shall be furnished at no cost to the Government, and that the manufacturer shall pay the transportation charges to and from the designated point where tests are to be made. In the case of failure of the sample or samples submitted, consideration will be given to the request of the manufacturer for additional tests only after it has been clearly shown that changes have been made in the product which the Government considers sufficient to warrant additional tests.

6.3 Definitions.-

6.3.1 Amplitude.- Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.3.2 Hermetic Seal.- A hermetic seal is defined as a perfectly closed and airtight seal made between vitric and/or metallic materials. A hermetic seal is not intended to include seals accomplished by gaskets.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Air Force

Other interest:
Navy - A

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

C-8279

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